Pediatric VF / Pulseless VT

PC-07 | Pediatric Cardiac

Quick flow

01 VF / Pulseless VT [EMT / AEMT / Paramedic]

Begin continuous CPR, effective BVM ventilation, and defibrillator/AED application. Minimize all interruptions.

02 Shock 1: 2 J/kg [AEMT / Paramedic]

Defibrillate, then immediately resume CPR for 2 minutes while preparing vascular access.

03 Shock 2: 4 J/kg [AEMT / Paramedic]

Defibrillate, resume CPR immediately, obtain IV/IO, and give epinephrine 0.01 mg/kg IV/IO.

04 Refractory Shockable Rhythm [AEMT / Paramedic]

Subsequent shocks: at least 4 J/kg, maximum 10 J/kg or adult dose. Give epinephrine every 3-5 minutes and consider one authorized antiarrhythmic.

05 Torsades / Reversible Cause? [Paramedic]

Paramedic: give magnesium only for torsades/polymorphic VT or suspected hypomagnesemia; identify and treat reversible causes.

06 ROSC? [EMT / AEMT / Paramedic]

Yes: PC-08 Pediatric Post-Resuscitation. No: continue 2-minute CPR, rhythm, shock, medication, and cause-treatment cycles.

Overview

- High-quality CPR and early defibrillation are the central treatments for pediatric VF/pulseless VT.
- Charge during compressions and resume CPR immediately after every shock without an immediate pulse check.

Provider-Level Actions

Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT

- Begin high-quality CPR and effective BVM ventilation, apply AED/monitor, minimize pauses, and resume compressions immediately after every shock.
- Use the pediatric length- or weight-based resuscitation system. Do not delay defibrillation for advanced procedures.

AEMT

- Perform all EMT care plus IV/IO access, epinephrine, and manual defibrillation only when authorized by current standing orders.
- Use IO promptly if IV access is not quickly available; do not delay CPR or defibrillation for repeated IV attempts.

Paramedic

- Perform all prior care plus manual defibrillation, one authorized antiarrhythmic for shock-refractory VF/pulseless VT, magnesium when indicated for torsades/polymorphic VT or suspected hypomagnesemia, $EtCO_2$-guided CPR improvement, and reversible-cause management.
- Charge during compressions, minimize the peri-shock pause, and do not interrupt compressions for endotracheal intubation.

Indications

- Pediatric pulseless arrest with ventricular fibrillation or pulseless ventricular tachycardia.

Treatment steps

- Use 2 J/kg for the first shock, 4 J/kg for the second shock, and at least 4 J/kg for subsequent shocks; do not exceed 10 J/kg or the adult dose.
- After an advanced airway is placed, use continuous compressions and ventilate 20-30/min. Use waveform $EtCO_2$ to improve CPR and identify possible ROSC, not as a stand-alone termination decision.
- Assess reversible causes during CPR: hypoxia, hypovolemia, acidosis, hypo/hyperkalemia or metabolic cause, hypothermia, tension pneumothorax, tamponade, toxins, and thrombosis.
- Routine calcium and sodium bicarbonate are not indicated unless a specific cause exists.
- After sustained ROSC, move immediately to PC-08 for controlled oxygenation/ventilation, BP and glucose reassessment, and destination notification.

Medications

Epinephrine 0.1 mg/mL (1:10,000)

0.01 mg/kg IV/IO after the second shock or sooner only if rapid defibrillation is not possible; maximum single dose 1 mg; repeat every 3-5 minutes

Amiodarone

5 mg/kg IV/IO for shock-refractory VF/pulseless VT; maximum 300 mg. Select one antiarrhythmic only unless Medical Control directs otherwise.

Lidocaine

1 mg/kg IV/IO for shock-refractory VF/pulseless VT; maximum 100 mg. Select one antiarrhythmic only unless Medical Control directs otherwise.

Magnesium sulfate

Paramedic only: 25-50 mg/kg IV/IO once; maximum 2 g; only for torsades/polymorphic VT or suspected hypomagnesemia

Warnings / precautions

- Do not pause CPR after a shock to check a pulse; resume immediately until the next planned rhythm check.
- Do not delay CPR or defibrillation for IV access, medication, or endotracheal intubation.
- Do not routinely administer calcium or sodium bicarbonate unless a specific indication exists.
- Use one antiarrhythmic only unless Medical Control directs otherwise.

Clinical pearls

- For initial shockable pediatric arrest, rapid defibrillation is the priority; give epinephrine after the second shock or sooner only when rapid defibrillation is not possible.
- Magnesium is not routine VF/pulseless VT treatment; reserve it for torsades/polymorphic VT or suspected hypomagnesemia.

Related pathways / contacts

AR-05 Pediatric Airway - Open for pediatric BVM and airway support.

AR-06 Pediatric Failed Airway - Open for ineffective oxygenation/ventilation or unsafe further airway attempts.

PC-04 Pediatric Pulseless Arrest - Open for overarching pediatric arrest routing.

PC-08 Pediatric Post-Resuscitation Care - Open immediately after ROSC.

Medication Reference - Open the current Claiborne medication reference.

Source and references

Claiborne County EMS approved protocol manual. | Current Tennessee EMS scope of practice and Claiborne County standing orders control for all provider permissions.

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